

Vector magnetometry with metastable strontium

Tensor control, probe-induced loss, and continuous readout

Discussion record based on the handwritten note of 26 September 2026

Status. This document records the model discussion and its consistency checks. It proposes a starting model; it does not report a new simulation or an agreed implementation. The existing v7 simulation is unchanged.

1 Physical proposal and atomic data

Prepare an initially uncorrelated ensemble of ^{88}Sr atoms in the metastable 3P_2 manifold, with electronic angular momentum $J = 2$ and five magnetic sublevels. An unknown magnetic field drives Larmor precession. A known magnetic control and the tensor light shift of a linearly polarized probe modify the spin dynamics. Faraday polarimetry of that same probe produces a continuous measurement record [1].

The probe is detuned from the $^3P_2 \rightarrow ^3D_1$ transition. Figure 1(c) of Ref. [2] shows the relevant levels and decay paths; Table II gives the following single-atom rates for ^{88}Sr .

Decay channel	Wavelength (nm)	Partial rate (s^{-1})
$^3D_1 \rightarrow ^3P_0$	2600	2.8×10^5
$^3D_1 \rightarrow ^3P_1$	2740	1.8×10^5
$^3D_1 \rightarrow ^3P_2$	3070	8.8×10^3

Summing these channels gives

$$\Gamma = 4.688 \times 10^5 \text{ s}^{-1}, \quad \frac{\Gamma}{2\pi} \simeq 74.6 \text{ kHz}, \quad (1)$$

$$b_{\text{return}} = \frac{8.8 \times 10^3}{\Gamma} \simeq 0.01877, \quad b_{\text{loss}} \simeq 0.98123. \quad (2)$$

Thus about 98.12% of scattering events leave the sensing manifold and 1.88% return. This supports the note’s loss-dominated approximation, with a small correction to its rounded statement of 99% loss. “Loss” means loss from 3P_2 , not necessarily physical escape from the trap. Treating these atoms as dark also assumes negligible repumping and negligible response of the other manifolds at the chosen probe frequency.

Scope of the paper comparison. We use the atomic structure and single-atom rates from Ref. [2]. The magnetometry model considered here assumes independent atoms, without the collective radiative interactions and superradiance studied in that paper.

2 Coherent dynamics and parameter conventions

Set $\hbar = 1$, so Hamiltonians have units of angular frequency. Let $\boldsymbol{\epsilon}$ be the real unit polarization vector of the probe and $\mathbf{k}$ its unit propagation direction. Define

$$J_\epsilon = \boldsymbol{\epsilon} \cdot \mathbf{J}, \quad J_k = \mathbf{k} \cdot \mathbf{J}, \quad \boldsymbol{\epsilon} \cdot \mathbf{k} = 0. \quad (3)$$

The tensor shift acts along the polarization axis; Faraday rotation measures the spin along the propagation axis. For example, keeping the current code's tensor term J_z^2 and readout J_y corresponds to $\boldsymbol{\epsilon} = \mathbf{e}_z$ and $\mathbf{k} = \mathbf{e}_y$. A readout of J_z instead requires a transverse polarization and the corresponding tensor axis.

Using the magnetic-field sign convention of the handwritten note,

$$H(t) = -\boldsymbol{\omega} \cdot \mathbf{J} + \Omega [\cos \phi(t) J_x + \sin \phi(t) J_y] + \frac{\beta}{2J} J_\epsilon^2, \quad J = 2. \quad (4)$$

Here $\boldsymbol{\omega}$ is the unknown Larmor-frequency vector, Ω is the known magnetic-control amplitude, and $\phi(t)$ is a prescribed control phase. The current v7 script uses $+\mathbf{w} \cdot \mathbf{J}$, so the conventions agree upon setting $\mathbf{w} = -\boldsymbol{\omega}$. This sign choice must be consistent in data generation, fitting, and conversion back to a magnetic field.

For an isolated, weakly excited optical transition, define

$$\Delta_L = \omega_L - \omega_{eg}, \quad V_0 = \frac{\Omega_L^2}{4\Delta_L}, \quad \gamma_0 = \frac{\Omega_L^2 \Gamma}{4\Delta_L^2}. \quad (5)$$

Ω_L is a reference optical Rabi frequency, distinct from Ω . For the angular factors below, it is normalized so that the actual absorption Rabi frequency equals Ω_L times the corresponding Clebsch–Gordan coefficient. All detunings and Rabi frequencies use angular-frequency units; Γ is the population decay rate in Eq. (1).

The relation to the simulation's tensor parameter is

$$\frac{\beta}{2J} = C^{(2)} V_0, \quad \frac{\beta/(2J)}{\gamma_0} = C^{(2)} \frac{\Delta_L}{\Gamma}. \quad (6)$$

For $C^{(2)} = -1/10$ and $J = 2$, this gives $\beta = -\Omega_L^2/(10\Delta_L)$, or equivalently $\beta/\gamma_0 = -(2/5)\Delta_L/\Gamma$. Here β is signed. v7 instead sweeps a positive magnitude and applies the tensor sign separately through its atomic preset.

Two normalization cautions. The coefficient multiplying J_ϵ^2 is $\beta/(2J)$, not β . Also, γ_0 is a reference scattering-rate scale, not the actual state-independent photon-scattering rate. The latter requires the angular operator derived on the next page. The note's later scaling argument suppresses these angular factors; they must be restored for quantitative comparisons.

3 State-dependent loss and the tensor correction

The handwritten coefficients are read as

$$C^{(0)} = \frac{1}{5}, \quad C^{(2)} = -\frac{1}{10}. \quad (7)$$

In the irreducible-tensor convention, the rank-two operator includes a trace subtraction [3]. Therefore the absorption-strength operator is

$$\begin{aligned} Q &= C^{(0)}\mathbb{I} + C^{(2)}\left(J_\epsilon^2 - \frac{J(J+1)}{3}\mathbb{I}\right) \\ &= \frac{1}{5}\mathbb{I} - \frac{1}{10}(J_\epsilon^2 - 2\mathbb{I}) = \boxed{\frac{4\mathbb{I} - J_\epsilon^2}{10}}. \end{aligned} \quad (8)$$

An independent angular-momentum check gives $Q_{mm} = |\langle 2, m; 1, 0 | 1, m \rangle|^2$. Explicitly,

m_ϵ	-2	-1	0	1	2
Q_{mm}	0	3/10	2/5	3/10	0

The literal expression $\mathbb{I}/5 - J_\epsilon^2/10$ in the note would give negative loss rates for $m_\epsilon = \pm 2$. The missing scalar contribution is essential. One may discard a common *real energy shift* from H , but cannot discard a common decay term from its non-Hermitian part.

Starting approximation: every scattering event is loss

Approximating the 98.12% loss branch by unity gives

$$K = \gamma_0 Q, \quad H_{\text{eff}} = H - \frac{i}{2}K, \quad \boxed{\dot{\rho} = -i[H, \rho] - \frac{1}{2}\{K, \rho\}}. \quad (9)$$

The density matrix here is unnormalized, with $\text{Tr } \rho(0) = 1$:

$$p(t) = \text{Tr } \rho(t), \quad \dot{p}(t) = -\gamma_0 \text{Tr}[Q\rho(t)]. \quad (10)$$

It describes the surviving part of the ensemble after ignoring the outgoing states. Since $Q \geq 0$, survival cannot increase. The instantaneous scattering rate per surviving atom is $\gamma_0 \text{Tr}(Q\rho/p)$ when $p > 0$.

Physical consequence. In the isolated-line approximation, $m_\epsilon = \pm 2$ are dark to the linearly polarized probe. Magnetic dynamics can move population between dark and bright states, so survival generally is not a single exponential independent of the unknown field and control.

Optional refinement: retain the small return branch

An explicit return model would add $\sum_a L_a \rho L_a^\dagger$ to Eq. (9), with $\sum_a L_a^\dagger L_a = b_{\text{return}} \gamma_0 Q$. The trace would then decrease at $b_{\text{loss}} \gamma_0 \text{Tr}(Q\rho)$. The L_a must be derived from the absorption and emission angular factors; v7's symmetric $J_\pm$ channels are not automatically the correct return model.

4 Measurement record, survival, and shot noise

Let N_0 be the initial atom number. Define the measurement strength κ operationally: in a time bin δt , the calibrated collective-spin readout has shot-noise variance $1/(\kappa\delta t)$. In the weak-backaction, shot-noise-dominated approximation, the record per initial atom is

$$m_i = \text{Tr}[J_k \rho(t_i)] + \frac{\xi_i}{N_0 \sqrt{\kappa \delta t}}, \quad \xi_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (11)$$

This treats each bin as sufficiently short to use its instantaneous signal; otherwise the mean must be averaged across the bin. It models ensemble means with additive photon shot noise, omitting resolved atom-number fluctuations, projection noise, and measurement-induced collective correlations.

Survival appears exactly once

Writing $\tilde{\rho} = \rho/p$ for the normalized state of surviving atoms,

$$\text{Tr}(J_k \rho) = p(t) \text{Tr}(J_k \tilde{\rho}). \quad (12)$$

No additional survival factor belongs in Eq. (11). Conversely, if we replace the loss operator by $K = \gamma_{\text{loss}} \mathbb{I}$, then

$$m_i = e^{-\gamma_{\text{loss}} t_i} \langle J_k \rangle_{\text{conditional}, i} + \frac{\xi_i}{N_0 \sqrt{\kappa \delta t}}. \quad (13)$$

Page 3 of the handwritten note includes the exponential in the collective record $M(t_i)$ but omits it in $m(t_i) = M(t_i)/N$. Dividing by N_0 does not remove survival. Dividing instead by the surviving number $N_0 p(t_i)$ would also amplify the noise by $1/p(t_i)$ and defines a different record.

Measurement strength needs a Sr-specific calibration

For a fixed isolated transition and geometry, the expected scaling is

$$\kappa = q_{\text{meas}} \frac{\sigma_{\text{ref}}}{A} \gamma_0, \quad \text{OD}_0 = \frac{N_0 \sigma_{\text{ref}}}{A}, \quad \text{Var}(m_i) = \frac{1}{q_{\text{meas}} N_0 \text{OD}_0 \gamma_0 \delta t}. \quad (14)$$

Here A is the beam mode area, σ_{ref} is a consistently defined reference cross section, and $q_{\text{meas}} > 0$ contains the vector polarizability, optical normalization, and detection efficiency. Its numerical value has not been derived in this discussion. Measurement strength involves the *square* of the Faraday signal coupling; the note's prefactor written as $C^{(1)}$ needs its convention specified.

The weak return branch also sets the dipole strength of the probed transition. A closed-two-level resonant cross section cannot be substituted without accounting for the branching and angular factors. The reference OD in Eq. (14) should not be confused with an off-resonant absorption optical depth, which itself depends on detuning and state.

For an initially uncorrelated spin coherent ensemble, an indicative weak-backaction condition is $\kappa T N_0 J/2 \ll 1$. Under Eq. (14), this becomes $q_{\text{meas}} \text{OD}_0 \gamma_0 T J/2 \ll 1$, up to the evolving spin variance. The note's $\text{OD} \gamma T \ll 1$ is the corresponding scaling estimate.

5 Detuning, resources, and the estimation objective

The proposed additional control is the detuning dependence of Eq. (6). Writing probe intensity as I_L , the isolated-line scalings are

$$|\beta| \propto \frac{I_L}{|\Delta_L|}, \quad \gamma_0 \propto \frac{I_L}{\Delta_L^2}, \quad \kappa \propto \gamma_0. \quad (15)$$

Consequently, the effect of increasing $|\Delta_L|$ depends on which resource is held fixed:

Fixed quantity	Required intensity	Effect of larger detuning
I_L	Constant	Weaker tensor dynamics, less scattering, weaker readout.
$ \beta $	$I_L \propto \Delta_L $	Less scattering, but also weaker readout.
γ_0	$I_L \propto \Delta_L^2$	Stronger tensor dynamics at fixed reference scattering and readout strength.

The last comparison directly tests whether extra nonlinear dynamics improves vector estimation at a fixed measurement-noise scale. With state-dependent loss, fixing γ_0 fixes the loss *operator scale*; the actual survival trajectory can still change as the Hamiltonian changes. Increasing the tensor strength does not by itself prove a Fisher-information gain.

These scalings require weak excitation and a sufficiently isolated transition, schematically

$$\Gamma, |\Omega_L| \ll |\Delta_L| \ll \text{separation to relevant neighboring optical lines}. \quad (16)$$

At larger detunings, contributions from other excited manifolds must be included. Available laser power and acceptable heating also constrain an experimental detuning sweep. Arbitrarily large detuning is not a consequence of this reduced model.

Accumulated information and useful measurement duration

For a fixed control and probe setting, let $\mu_i(\omega) = \text{Tr}[J_k \rho(t_i; \omega)]$ and let the known shot-noise variance be σ_i^2 , independent of the unknown field. Then

$$F(T) = \sum_{t_i \leq T} \frac{1}{\sigma_i^2} (\nabla_{\omega} \mu_i) (\nabla_{\omega} \mu_i)^{\top}. \quad (17)$$

Each additional measurement contributes a positive semidefinite matrix. Accumulated information cannot decrease when a record is extended under otherwise identical conditions, although signal loss may make the gain very small. Where F is invertible, the associated Cramer–Rao bound cannot worsen.

A finite optimal duration can arise when optimizing information per total experimental time, including repeated preparation and dead time, or when jointly changing the probe settings at fixed duration. Thus $\gamma_{\text{loss}} T \sim 1$ is a useful heuristic timescale for uniform loss, not a general maximum of accumulated Fisher information. For vector estimation, a scalar objective such as $\text{Tr } F^{-1}$ must also be specified.

6 Comparison with v7 and proposed next decisions

The current script is `multiparameter_trotter_v7_loss_least_squares_over_time.py`. Its loss generator already contains a term $-\{K_L, \rho\}/2$, with $K_L = \gamma_s(a_L\mathbb{I} + c_L J_z^2)$, alongside within-manifold spin-flip channels. Thus the trace-decreasing formalism can be reused; the atomic coefficients, probe parametrization, and any return channels require a new model.

Ingredient	Current v7	Proposed Sr model
Sensing manifold	Cs $F = 3$ or 4	Sr $J = 2$
Irreversible dynamics	Spin flips and Raman loss	Dominant loss; optional 1.88% return branch
Tensor/scattering ratio	Fixed atomic preset	Detuning-dependent ratio
Probe parameters	Sweep positive β magnitude	Intensity and detuning set shift, scattering, and noise
Loss geometry	Preset $a_L + c_L J_z^2$	$Q = (4\mathbb{I} - J_\epsilon^2)/10$
Readout	Unnormalized signal plus shot noise	Same signal convention; Sr-specific optical calibration needed

Recommended starting point, not yet an agreed choice. Use Eqs. (4), (5), (6), (8), (9), and (11). Retain state-dependent loss while initially treating all scattering as loss. Compare it with a uniform-loss model to determine whether state-selective survival changes the inferred metrological advantage. The five-dimensional spin space makes this comparison modest in size.

The remaining decisions are:

- Retain state-dependent loss immediately, or start with a scalar approximation? If scalar, specify how its rate is calibrated: the isotropic average is $\text{Tr } Q/5 = 1/5$, while an initial spin coherent state transverse to ϵ has $\langle Q \rangle = 3/10$.
- Choose polarization, propagation, initial spin, and control axes together.
- Derive the Faraday coefficient and cross-section normalization before assigning a numerical OD or shot-noise level.
- Specify the detuning/power constraint and the Fisher-information objective for the first comparison. Add the small return branch only when its effect is needed at the desired accuracy.

References

- [1] IHD, handwritten note, *Vector Magnetometry*, September 2026, pp. 1–3 (page 4 is blank). Repository file: `Vector Magnetometry - 20260926_151513.pdf`.
- [2] S. J. Masson, J. P. Covey, S. Will, and A. Asenjo-Garcia, *Dicke Superradiance in Ordered Arrays of Multilevel Atoms*, PRX Quantum **5**, 010344 (2024), Fig. 1(c) and Table II. doi:10.1103/PRXQuantum.5.010344. Repository file: `PRXQuantum.5.010344.pdf`.
- [3] I. H. Deutsch and P. S. Jessen, *Quantum control and measurement of atomic spins in polarization spectroscopy*, Optics Communications **283**, 681–694 (2010). See Eq. (14) for the tensor convention and Sec. 3.2.1 for measurement strength. doi:10.1016/j.optcom.2009.10.059.